



**CALICUT UNIVERSITY – FOUR-YEAR
UNDER GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	APPLIED NUCLEAR PHYSICS				
Type of Course	Major Elective (SPECIALIZATION III: PHYSICS IN BIOLOGY)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Fundamental ideas in mechanics, electromagnetism, and mathematical physics along with the basic understanding of concepts in modern physics like atomic and nuclear structure.				
Course Summary	The course in Applied Nuclear Physics provides an in-depth account of the fundamental constituents of matter, their interactions, and the underlying principles governing nuclear structure, particle behaviour, and their implications in different walks of modern technology.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand Radioactive Processes: Explain the mechanisms and types of radioactive decay. Understand internal conversion and their roles in radioactive decay chains and environmental radioactivity.	Understand	Conceptual Knowledge	Quizzes, Tests

CO2	Analyze Nuclear Collisions and Reactions: Describe nuclear collision processes, understand nomenclature and probes, calculate cross sections and reaction rates, and discuss examples of isotope production and nuclear reactions, including elastic scattering and resonance.	Analyse	Procedural Knowledge	Homework Assignments
CO3	Apply Radiation Interaction Principles: Utilize the Bethe-Bloch formula to predict energy loss of heavy charged particles in matter, interpret Bragg curves, and analyze the dependence on projectile and medium. Understand gamma-ray attenuation and neutron interaction processes including attenuation and moderation.	Apply	Conceptual Knowledge	Problem Sets, Projects
CO4	Explore Neutron Physics: Discuss the properties of neutrons, classify different types of neutrons, and understand the various sources of neutrons. Use neutron detectors like the BF3 counter.	Analyse	Procedural Knowledge	Homework, Exams
CO5	Assess Biological Effects of Radiation: Evaluate the biological impacts of radiation exposure, differentiate between direct and indirect physical and chemical damage, calculate dose and dose rate, and understand dose distribution and its relative biological effectiveness. Assess human exposure from natural and artificial sources.	Understand	Basic Concepts	Virtual lab Demonstrations
CO6	Utilize Radiation in Industrial and Analytical Applications: Demonstrate the use of radiation in industrial applications. Apply analytical techniques for materials analysis.	Apply	Conceptual Knowledge	Problem Sets, Projects
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	RADIOACTIVITY AND NUCLEAR COLLISIONS		12	20
	1	Radioactivity and Radioactive Decay – Alpha emission, beta emission and electron capture, gamma emission and internal conversion	2	
	2	Rate of radioactive decay, Radioactive decay chains	2	
	3	Radio activity in the environment, Radioactive dating	1	
	4	Nuclear collisions, Nomenclature, Probes	1	
	5	Cross sections, differential cross- section and reaction rates	2	
	6	Isotope production, examples of nuclear reactions	2	
	7	Elastic scattering, Resonance. (Sufficient exercises)	2	
	Sections 1.5 and 1.6 of Book 1			
II	RADIATION INTERACTION AND NEUTRON PHYSICS		12	20
	8	Interaction of Radiation with Matter – Heavy charged particles	2	
	9	Bethe-Bloch Formula, Energy Dependence	2	
	10	Bragg curve, Projectile dependence, Medium dependence	2	
	11	Gamma ray attenuation, Neutrons, attenuation, neutron moderation	2	
	12	Neutron Physics - Discovery of neutron, Properties of neutrons, Magnetic moment measurement, Classification of neutrons	2	
	13	Sources of neutrons, radioactive sources, photo neutrons, Neutron detector-(BF ₃ counter) (Sufficient exercises)	2	
	Sections from 5.1 and 5.5 (Book for study 1) and 13.1 to 13.11 (Book 2)			
III	BIOLOGICAL EFFECTS OF RADIATIONS		12	15

	14	Biological effects of radiations - direct and indirect physical damage, indirect chemical damage	2	
	15	Dose, Dose rate and Dose distribution	3	
	16	Dose distribution and relative biological effectiveness, equivalent and effective dose	2	
	17	Damage to critical tissues	2	
	18	Human exposure to radiation, Natural sources, Artificial sources of exposure. (Sufficient exercises)	3	
	Section 7.1 and 7.5.2 of Book 1			
IV	INDUSTRIAL AND ANALYTICAL APPLICATIONS		12	15
	19	Industrial and Analytical Applications – Industrial uses, Tracing, Gauging	2	
	20	Material modification, sterilization, Neutron Activation Analysis	3	
	21	Rutherford Back scattering	2	
	22	Particle Induced x-ray emission techniques for materials analysis	3	
	23	Accelerator mass spectrometry. (Sufficient exercises).	2	
	Sections from 8.1 to 8.6 of Book 1			
V	OPEN-ENDED MODULE NUCLEAR MEDICINE - X-radiography and gamma camera, Positron Emission Tomography, Radiation therapy- using photons and electrons, radio nuclides, Neutron therapy, therapy with heavy charged particles, (Book 1) Or REACTOR PHYSICS – Neutron moderation mechanism, slowing down of neutrons, moderation ratio, diffusion of neutrons, reactor equation, critical size. (Book 2) COMMERCIAL NUCLEAR REACTORS – Gas cooled reactors, Pressurized water reactor, Heavy water reactors, Breeder reactors. THERMONUCLEAR FUSION – Basic principles, fusion containers –magnetic confinement, inertial confinement Or USE OF PYTHON AS AN ANALYTIC TOOL - Radiation attenuation in different material media can be modelled, computed and plotted with Python		12	

	Problems on material analysis in NAA, RBS and PIXE etc. can be done with Python programming. Analysis of Neutron diffusion and moderation mechanisms with appropriate computational tools.		
Books and References: 1. Nuclear Physics - Principles and Applications, John Lilley, Manchester Physics series, John Wiley and Sons (Book 1) 2. Nuclear Physics – SN Ghoshal, S-Chand & Company(Book 2) Supplementary Readings - 3. Atomic and Nuclear Physics, N. Subrahmanyam , Brij Lal , Jivan Seshan, , S-Chand and company 4. Nuclear Physics, Anwar Kamal, Springer Publishers, 5. Nuclear Physics, D. C. Tayal, Himalaya Publishing House 6. The Basics of Nuclear Physics, Christopher Cooper, Roshan Publishing group, NY. 7. Nuclear Methods in Science and Technology, Yuri M. Tsipenyuk, IOP Publications			

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	3	2	0	2	2	1	2	2	0	2	2
CO 2	3	2	2	3	2	0	3	0	0	2	2	0	0
CO 3	3	3	3	2	0	2	2	2	1	0	2	1	2
CO 4	2	2	2	2	0	0	3	0	2	2	0	0	0
CO 5	2	3	2	2	1	3	1	2	0	2	1	2	2
CO 6	0	2	2	2	0	2	0	0	2	0	0	2	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	